

DATA STRUCTURE

LAB PROGRAMS

19. Heap Sort

```
#include <stdio.h>

void heapify(int arr[], int n, int i) {
    int largest = i, left = 2*i+1, right = 2*i+2;
    if(left < n && arr[left] > arr[largest])
        largest = left;
    if(right < n && arr[right] > arr[largest])
        largest = right;
    if(largest != i) {
        int temp = arr[i];
        arr[i] = arr[largest];
        arr[largest] = temp;
        heapify(arr, n, largest);
    }
}

void heapSort(int arr[], int n) {
    for(int i = n/2-1; i >= 0; i--)
        heapify(arr, n, i);
    for(int i = n-1; i > 0; i--) {
        int temp = arr[0];
        arr[0] = arr[i];
        arr[i] = temp;
        heapify(arr, i, 0);
    }
}
```

```
}  
  
int main() {  
    int arr[] = {23,56,12,34,78,29};  
    int n = 6;  
    heapSort(arr,n);  
    printf("Sorted Array:\n");  
    for(int i=0;i<n;i++)  
        printf("%d ",arr[i]);  
    return 0;  
}
```

Output

Sorted Array:

12 23 29 34 56 78

=== Code Execution Successful ===